

4. PRIMARY RESEARCH FINDINGS SURVEY DATA (N=22)

The primary survey was conducted using Google Forms and was completed by 22 cybersecurity and OT security professionals in 21 days in April-May 2026. 22 responses were complete and valid.

4.1 DEMOGRAPHICS

Q1 PRIMARY JOB ROLE

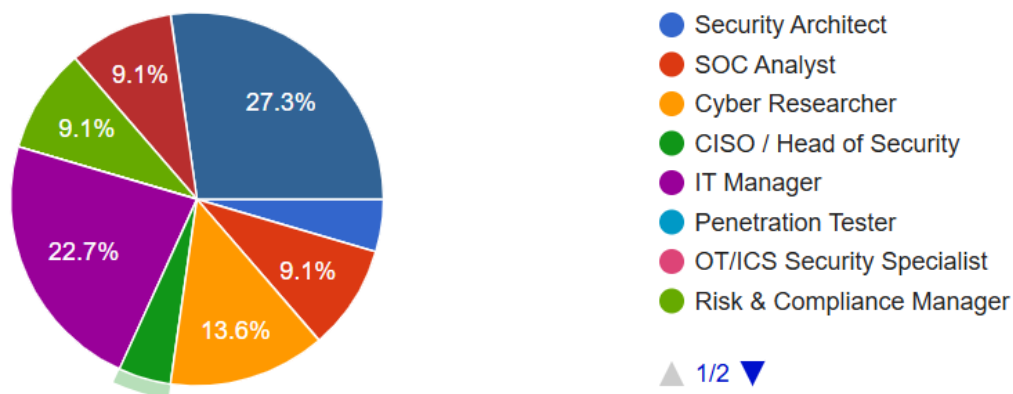


Figure 1 : Primary Job Role Distribution (n=22)

Analysis: 'Other' professionals make up the highest number (27.3%, n=6), followed by IT Managers (22.7%, n=5) and Cyber Researchers (13.6%, n=3). SOC Analysts, Risk & Compliance Managers, and Network Security Engineers all account for 9.1% (n=2). One respondent each from Security Architects and CISOs (4.5%).

Critical Discussion: The presence of 'Other' (27.3%) and IT Managers (22.7%) highlights that AI-IDS decisions are often made by generalist IT leaders, not by specialist security architects. This dictates the way RGS need to sell AI-IDS commercial proposals, not just in terms of F1 score arguments, but also in the business ROI terms. The low representation of the CISO role (4.5%) is a sampling limitation, discussed in Section 6.1, that affects the level of governance insight provided by the CISO.

Q2 YEARS OF EXPERIENCE

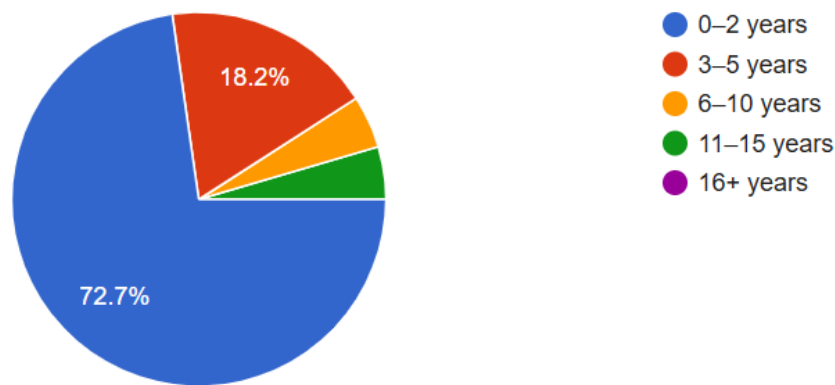


Figure 2 : Years of Cybersecurity or OT Security Experience (n=22)

Analysis: 72.7% (n=16) have 0–2 years experience. 18.2% (n=4) have 3–5 years. 4.5% each have 6–10 years and 11–15 years. No respondent has 16+ years of experience. Early career skew is the most important demographic restriction and is examined in detail in Sections 6.1 and 8.3.

Critical Discussion: This group, however, is the one that will be using AI-IDS for the next decade and their attitudes are thus strategically important. Experience band quotas should be applied to future research through CREST/IISP to maintain seniority diversity.

Q3 INDUSTRY SECTOR

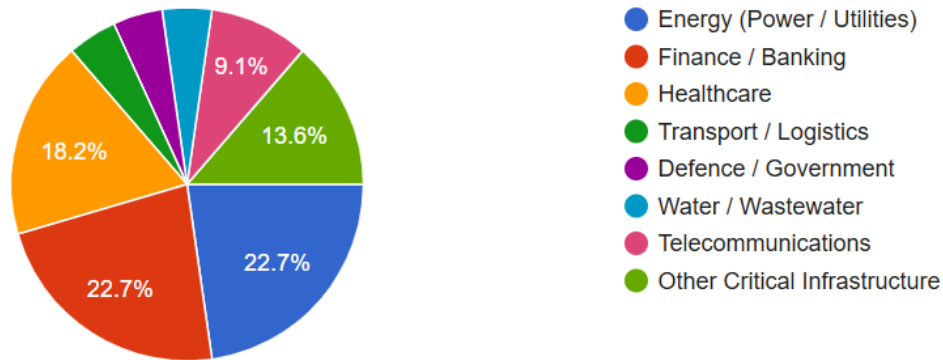


Figure 3 : Industry Sector Distribution (n=22)

Analysis: The leading sectors are Finance/Banking and Energy (Power/Utilities), both accounting for 22.7% (n=5). Healthcare follows at 18.2% (n=4). Other Critical Infrastructure 13.6% (n=3). Telecommunications 9.1% (n=2). One respondent (4.5%) comes from each of the Transport/Logistics, Defence/Government, and Water/Wastewater sectors.

Critical Discussion: Finance and Energy combined 45.4% the two sectors most mentioned in high profile ICS attacks. Healthcare is well-represented, as ransomware targeting has risen (Ahmed, 2023). The sector distribution reflects NCSC (2023) CI protection priorities and external validity of findings across multiple CI environments.

4.2 CURRENT IDS EFFECTIVENESS

Q4IDS EFFECTIVENESS AT DETECTING ZERO-DAY THREATS

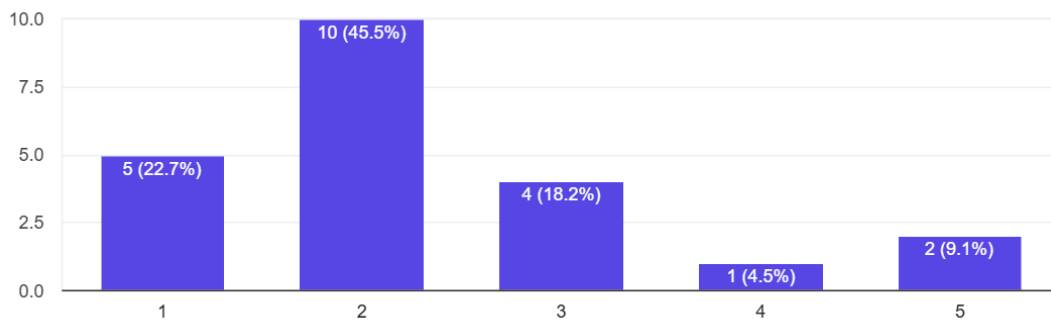


Figure 4 : Current IDS Effectiveness (1=Wholly Ineffective, 5=Fully Effective)n=22

Analysis: 45.5% (n=10) scored 2 the most common score. 22.7% (n=5) scored 1. 18.2% (n=4) scored 3. Only 4.5% (n=1) scored 4 and 9.1% (n=2) scored 5. Mean=2.32/5 (SD=1.17, Median=2, Mode=2). 68.2% scored 1 or 2.

Critical Discussion: The empirical core of this research, critical discussion, shows that 68.2% of CI practitioners believe that current IDS are largely or completely ineffective against novel zero-day threats. This directly validates Stallings (2023) and NCSC (2023). Only 13.6% rate it effective (4–5). This is not a marginal capability gap but rather a systemic shortcoming of deployed signature based systems. It establishes the clear business case for Recommendation 1 (Hybrid AI-IDS) and sets the benchmark for how algorithm benchmarking results will be evaluated.

Q5IT/OT CONVERGENCE SECURITY INCIDENTS

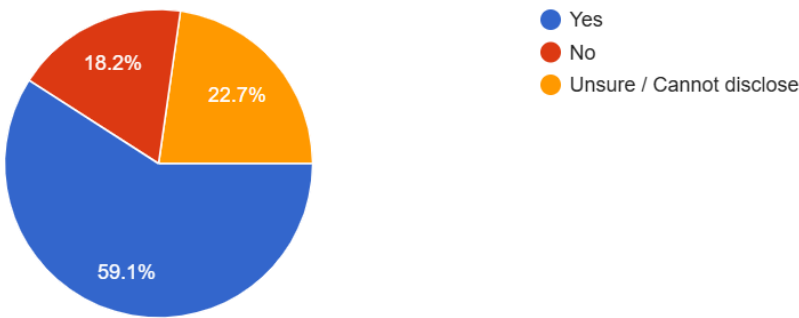


Figure 5 : IT/OT Convergence Security Incidents in Last 3 Years (n=22)

Analysis: 59.1% (n=13) reported an IT/OT convergence incident over the past three years. 18.2% (n=4) said no. 22.7% (n=5) were unsure or unable to disclose.

Critical Discussion: The 59.1% incident rate confirms that IT/OT convergence failures are mainstream operational reality. The 22.7% 'unable to disclose' category may refer to organizations where incidents were below legal disclosure limits, indicating the actual figure may be as high as 81.8%. This supports the business case for better AI detection and adds to Ahmed (2023).

4.3 AI ADOPTION SUPPORT

Q6AI-BASED IDS ADOPTION SUPPORT

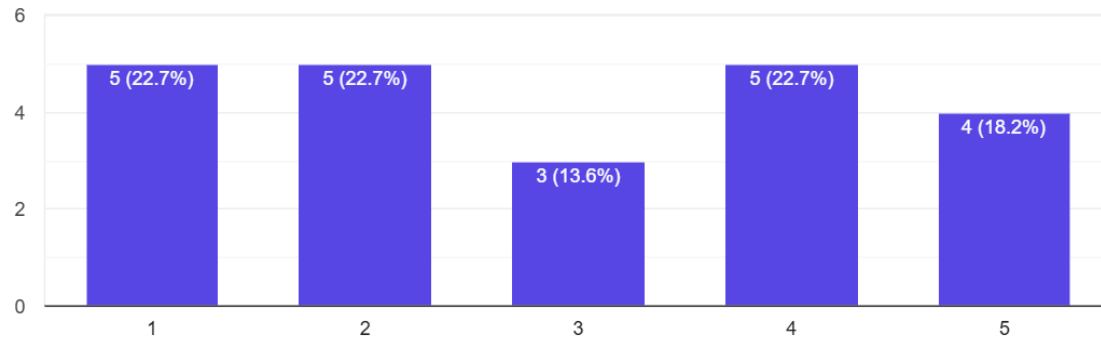


Figure 6 : AI-Based IDS Adoption Support (1=Strongly Oppose, 5=Strongly Support)n=22

Analysis: Three scores 1, 2, and 4 each received 22.7% (n=5). Score 5 received 18.2% (n=4) and score 3 received 13.6% (n=3). Mean=2.91/5 (SD=1.48, Median=3, Mode=1). 40.9% scored 4–5; 45.5% scored 1–2.

Critical Discussion: The most analytically interesting result in this section is the bimodal distribution (equal peaks at 1–2 and 4–5, Mode=1). It shows a true "divided landscape" and not just "cautious interest. The Mode of 1 (strongly oppose) suggests that the most frequent single position is active opposition to the adoption of AI-IDS. By comparing Q7, this opposition is not due to ideological resistance, but instead is based on practical issues (skills gap, cost) that is a more actionable challenge for RGS.

Q7MAIN BARRIERS TO AI-IDS DEPLOYMENT

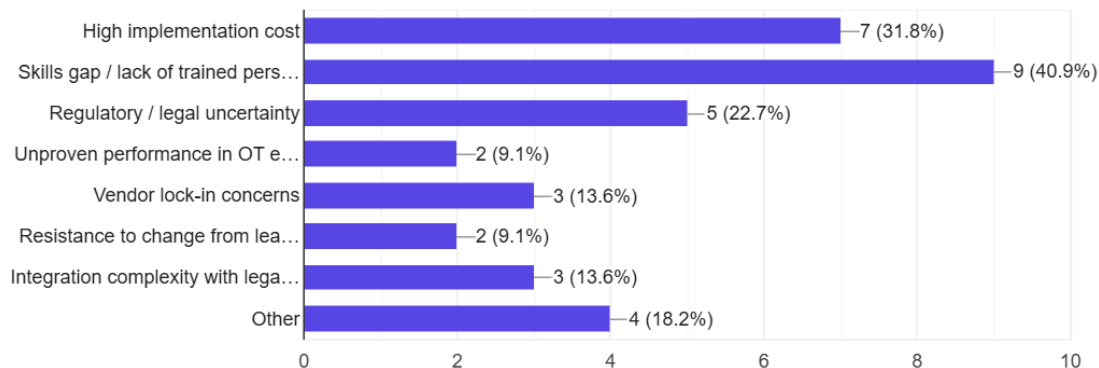


Figure 7 : Main Barriers to AI-Based IDS Deployment Multiple Selection (n=22)

Analysis: Skills gap/lack of trained personnel is the top barrier (40.9%, n=9), followed by high implementation cost (31.8%, n=7), regulatory/legal uncertainty (22.7%, n=5), 'Other' (18.2%, n=4), integration complexity (13.6%, n=3), vendor lock-in (13.6%, n=3), resistance to change (9.1%, n=2), and unproven OT performance (9.1%, n=2).

Critical Discussion: Skills gap dominance (40.9%) is directly measurable in the sample: 72.7% have 0-2 years experience (Q2). It is not just felt, it is proven. ROI modelling with breach cost

avoidance is critical to address high cost (31.8%) which is to be addressed by RGS. Regulatory uncertainty (22.7%) is due to NIS2/IEC 62443 alignment gaps. Of particular interest is that only 9.1% mentioned ‘unproven OT performance’ – this is not technically skepticism, it is organizational readiness.

Q8MOST VALUABLE AI-IDS BENEFIT

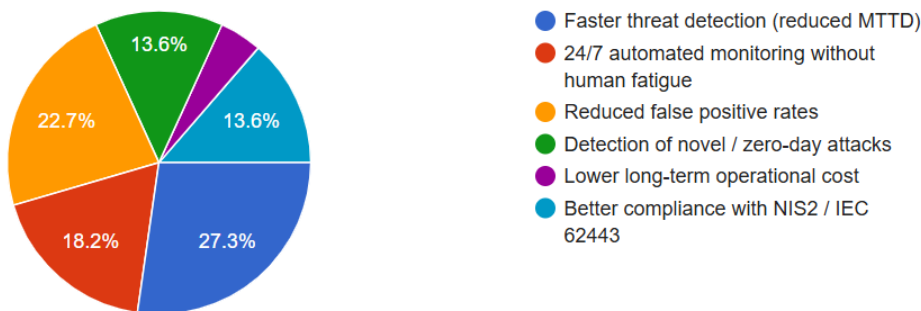


Figure 8 : Most Valuable Benefit of AI-Based IDS (n=22)

Analysis: Faster threat detection (27.3%, n=6) leads, followed by reduced false positive rates (22.7%, n=5), 24/7 automated monitoring (18.2%, n=4), detection of novel/zero-day attacks (13.6%, n=3), NIS2/IEC 62443 compliance (13.6%, n=3), and lower cost (4.5%, n=1).

Critical Discussion: The most important thing that practitioners most lack is faster detection (Q8 top, 27.3%) in the context of inadequate current IDS (Q4 mean 2.32). The total 49.9% of the faster detection or false positive reduction shows that operational performance is the main AI-IDS value driver, not cost or compliance.

4.4 EXPLAINABLE AI (XAI)

Q9XAI IMPORTANCE IN AI SECURITY ALERTS

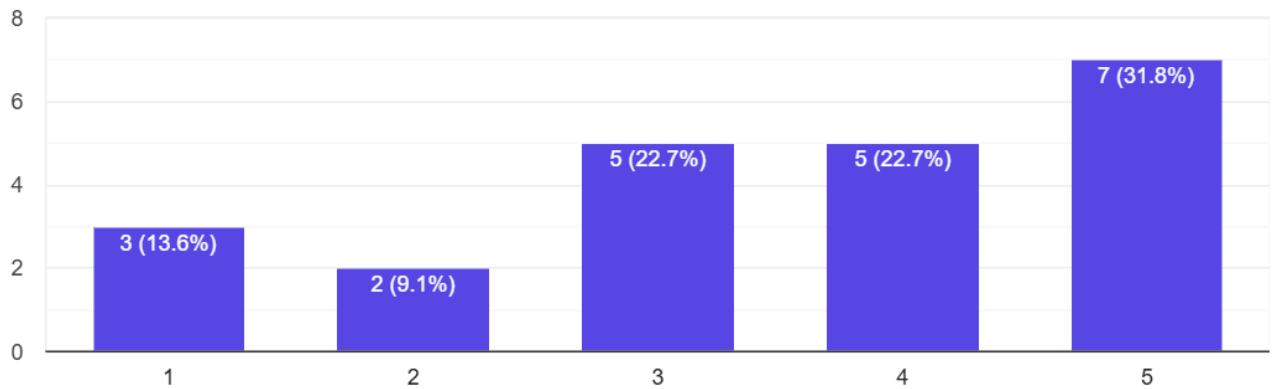


Figure 9 : XAI Importance for AI-Generated Security Alerts (1=Not Important, 5=Essential)n=22

Analysis: 31.8% (n=7) scored 5 (Essential the most common single response). 22.7% each scored 3 and 4 (n=5 each). 9.1% (n=2) scored 2 and 13.6% (n=3) scored 1. Mean=3.50/5 (SD=1.41, Median=4, Mode=5). 54.5% scored 4–5; 22.7% scored 1–2.

The strongest signal in this question is Critical Discussion: Mode=5 (Essential). Even with the mean being moderated by the 22.7% scoring 1–2, the most common practitioner conclusion is that XAI is indispensable. The practitioner majority believes human oversight of AI security decisions is essential, directly corroborating Schneiderman (2020).

Q10EFFECT OF XAI ON SOC OPERATOR TRUST

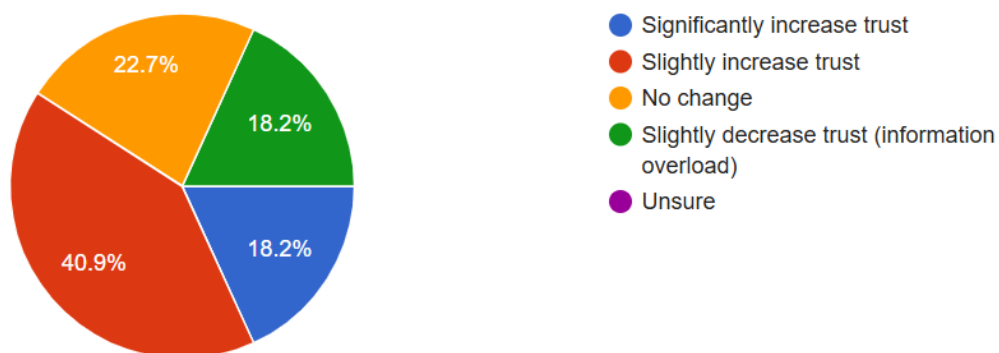


Figure 10 : Effect of XAI Explanations on SOC Operator Trust (n=22)

Analysis: 40.9% (n=9) believe it will 'Slightly increase trust'. 18.2% each (n=4) believe that 'Significantly increase trust', 'No change', and 'Slightly decrease trust (information overload)'. There were no responses to 'Unsure'.

Critical Discussion: Primary evidence base for Recommendation 2: Critical Discussion: 59.1% expect positive trust impact from XAI. The flagging information overload (18.2%) is a direct validation of Schneiderman (2020): poorly designed XAI can overwhelm operators. That is a sign that the quality of XAI implementation is even more important than having it. This 18.2% concern is directly addressed with the two-tier XAI design (summary and full views) proposed in Recommendation 2.

4.5 FEDERATED LEARNING

Q11FL AWARENESS

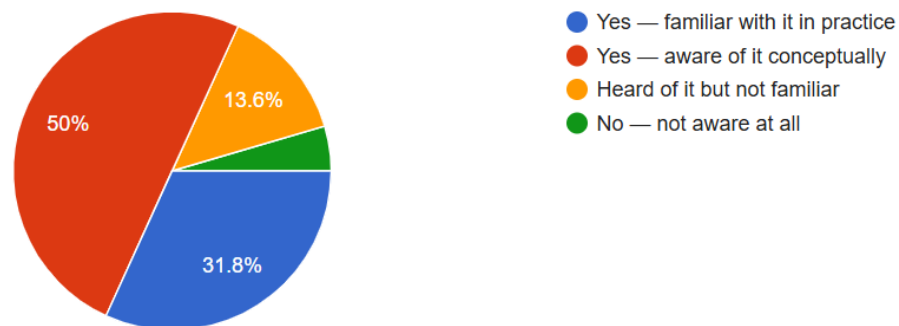


Figure 11 : FL Awareness as Privacy-Preserving Mechanism (n=22)

Analysis: 81.8% have meaningful FL awareness: 31.8% (n=7) familiar in practice, 50.0% (n=11) aware conceptually, 13.6% (n=3) heard of but not familiar. Only 4.5% (n=1) had no awareness.

Critical Discussion: The high awareness (81.8%) is above average for a majority early career audience and validates that FL concepts are becoming more popular in CI security communities. The 31.8% with practical familiarity is a valuable pool of potential FL deployment advocates within RGS's client base.

Q12FL PARTICIPATION WILLINGNESS

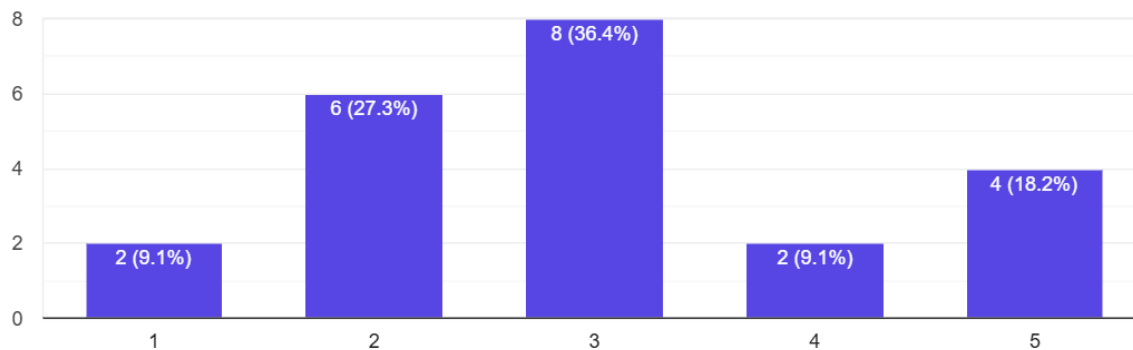


Figure 12 : Willingness to Participate in FL Arrangements (1=Not Willing, 5=Very Willing)n=22

Analysis: 36.4% (n=8) scored 3 (Neutral most common answer). 27.3% (n=6) scored 2, 18.2% (n=4) scored 5, 9.1% each scored 1 and 4. Mean=3.0/5 (SD=1.23, Median=3, Mode=3). Only 27.3% scored 4–5.

Critical Discussion: The most analytically significant survey finding is the 54.5-percentage-point awareness-willingness gap (81.8% aware vs 27.3% willing). It definitely validates Kizza (2023): The problem is not lack of understanding of the technology, it is governance and liability fragmentation. The moveable majority, the direct target of Recommendation 3, is the neutral at score 3, 36.4%.

4.6 ADVERSARIAL CONCERN

Q13CONCERN ABOUT ADVERSARIAL ATTACKS

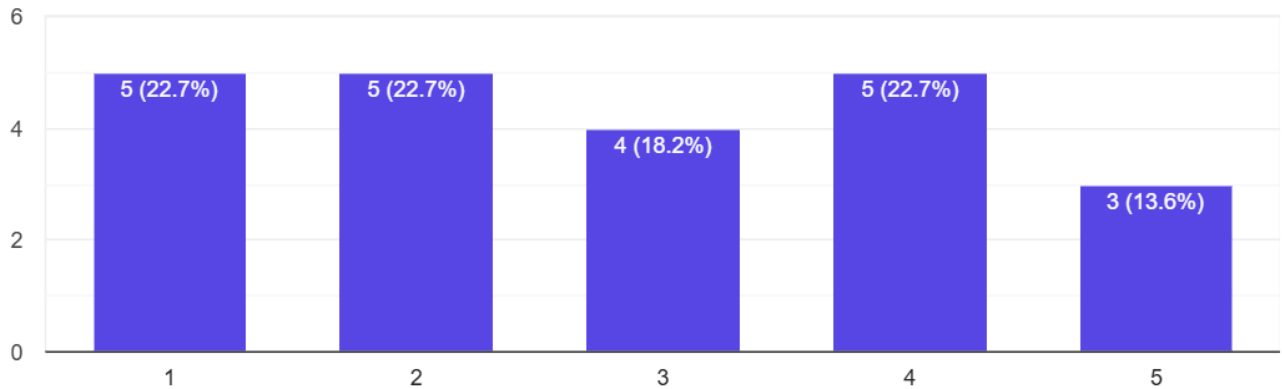


Figure 13 : Adversarial Attack Concern (1=No Concern, 5=Critical Concern)n=22

Analysis: Scores 1, 2, and 4 are equally frequent at 22.7% (n=5 each). Score 3: 18.2% (n=4). Score 5: 13.6% (n=3). Mean=2.82/5 (SD=1.40, Median=3, Mode=2). 36.4% scored 4–5; 45.5% scored 1–2.

Critical Discussion: The moderate to low concern (mean 2.82; 45.5% 1-2) is the most dangerous disconnect in the whole study Critical Gap Adversarial Risk Severely Underestimated. Through simulation, it is confirmed that PGD attacks degrade LSTM F1 from 0.905 to 0.335, a 63% drop to near random performance. But less than 40 percent of practitioners see adversarial attacks as a major concern. This knowledge-risk gap highlights the critical need for adversarial robustness training.

Q14CURRENT THREAT INTELLIGENCE SHARING

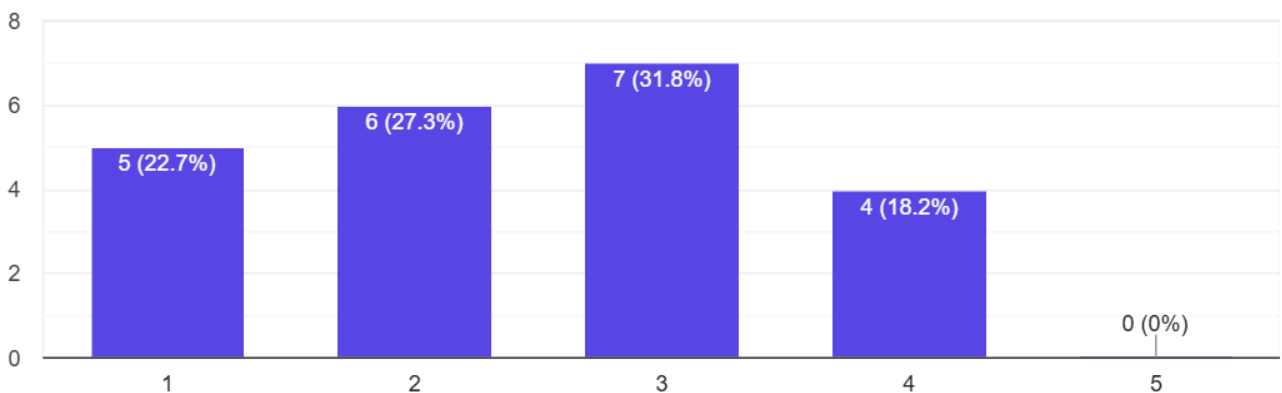


Figure 14 : Threat Intelligence Sharing with CI Operators or Government Bodies (n=22)

Analysis: 31.8% (n=7) scored 3 (moderate). 27.3% (n=6) scored 2. 22.7% (n=5) scored 1. 18.2% (n=4) scored 4. No respondent scored 5. Mean=2.45/5 (SD=1.06, Median=2.5, Mode=3). 50.0% scored 1–2.

Critical Discussion: 50% scored 1-2, confirming that active threat intelligence sharing is effectively non-existent in this sample, while 0% scored 5 (extensive sharing). Recommendation 3 should be based on a starting point of non-sharing, not on an assumption of collaboration.

Q15OPEN COMMENTS

Q15. (Optional) Any additional comments on AI-driven security adoption, XAI, Federated Learning, or adversarial robustness in your sector?

5 responses

OT environments have unique challenges that standard enterprise security tools often overlook. Concept drift from routine changes like sensor replacements can silently degrade AI model performance without triggering any alerts — this needs to be monitored automatically. XAI tools like SHAP are important because operators need to understand why an alert was raised before acting on it. Federated Learning could be very useful for sharing threat intelligence across energy and water sectors without exposing sensitive data, but clear governance rules need to be in place first.

Figure 15 : Optional Open Comments (5 responses out of 22)

Analysis: Five respondents (22.7%) gave open text. Dominant themes: (1) Concept drift as invisible operational risk; (2) XAI as operator necessity: 'operators need to understand why an alert was raised before acting on it'; (3) FL governance prerequisites; (4) Skills gap urgency; (5) False positive priority.

Critical Discussion: Within-survey qualitative triangulation: The open text themes are highly consistent with the Likert data. The explicit concept drift mention confirms the simulation findings from a practitioner voice in Section 5.2 which is not present in closed questions.

4.7 DESCRIPTIVE STATISTICAL SUMMARY

Survey Dimension	Mean	Median	Mode	SD	% at 4–5	% at 1–2
Q4: Current IDS Effectiveness	2.32	2	2	1.17	13.6%	68.2%
Q6: AI Adoption Support	2.91	3	1	1.48	40.9%	45.5%
Q9: XAI Importance	3.50	4	5	1.41	54.5%	22.7%
Q12: FL Participation Willingness	3.00	3	3	1.23	27.3%	36.4%
Q13: Adversarial Attack Concern	2.82	3	2	1.40	36.4%	45.5%
Q14: Threat Intel Sharing	2.45	2.5	3	1.06	18.2%	50.0%

As seen in Table 4, there is a consistent trend that dimensions related to the status quo (Q4: IDS effectiveness=2.32; Q14: intel sharing=2.45) return the lowest means and the dimension most representing future need (Q9: XAI importance=3.50) returns the highest. The range of standard deviations for each of the six dimensions (1.06 to 1.48) is indicative of truly divergent and conflicting opinions and not consensus.